

STaB-RAG: Weighted Multi-Signal Retrieval for Semi-Structured Tables

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Abstract

Current Retrieval-Augmented Generation (RAG) systems often linearize tables and rely on a single similarity score, discarding structural cues that are crucial for table question answering. On the NQ-table benchmark, however, a standard DPR baseline already reaches 95% Recall@50, indicating that existing methods usually retrieve the correct table but frequently fail to rank it highly, precisely the step that matters for accurate LLM generation. We introduce **STaB** (Structure-Text-and-BM25) RAG, a weighted reranking pipeline that fuses dense similarity, lexical overlap, and structural alignment signals to reorder retrieved tables. Experiments on table QA benchmarks show that this STaB-RAG setup improves ranking quality over dense-only and lexical-only baselines, leading to more accurate and interpretable generations. Our full implementation is available at: github.com/Abhijit85/WeightedRAG.

1 Problem Statement

Semi-structured tables support a growing share of real-world knowledge, from financial and healthcare records to sports statistics and social media metadata (Pasupat and Liang, 2015; Chen et al., 2020), making precise retrieval and reasoning essential. While large language models excel at language, they often hallucinate and rely on static, parametric knowledge; Retrieval-Augmented Generation (RAG) mitigates this by pulling in external evidence (Lewis et al., 2020; Guu et al., 2020). However, prevailing RAG pipelines flatten tables into plain text, underweight schema and metadata, and rely on generic rerankers that ignore table context such as headers, column types, and layout (Herzig et al., 2020; Yin et al., 2020; Yu et al., 2025).

On the NQ-table benchmark, a standard DPR setup already attains around 95% Recall@50

(Karpukhin et al., 2020; Kwiatkowski et al., 2019), indicating that the correct table is usually present somewhere in the retrieved pool but often fails to appear in the top ranks where it can actually guide the language model’s answer. This reveals a specific gap: for semi-structured tables, the central challenge is not merely retrieving some relevant tables, but designing retrieval and ranking objectives that are explicitly sensitive to tabular structure, schema, and domain signals so that the most informative table evidence is consistently surfaced.

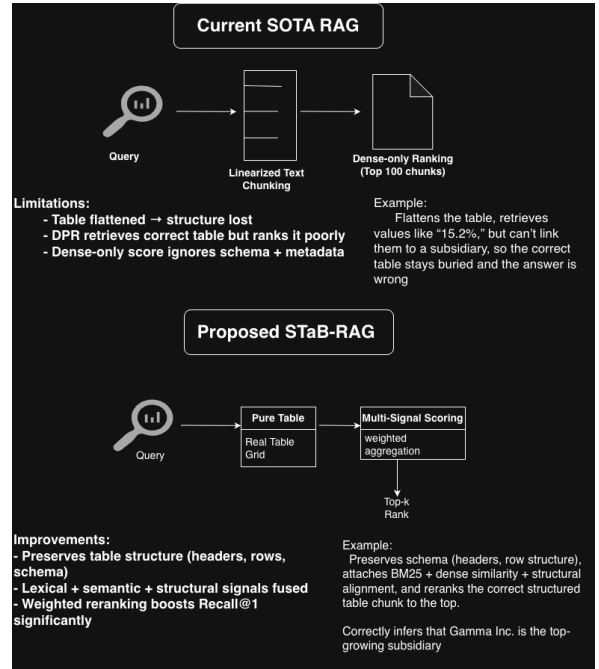


Figure 1: Problem Statement

2 Approach

2.1 Phase 1: Construct NQ-table BEIR Corpus

We begin with the Natural Questions (NQ) dataset (Kwiatkowski et al., 2019), extracting all table-based QA instances from the raw HTML and anno-

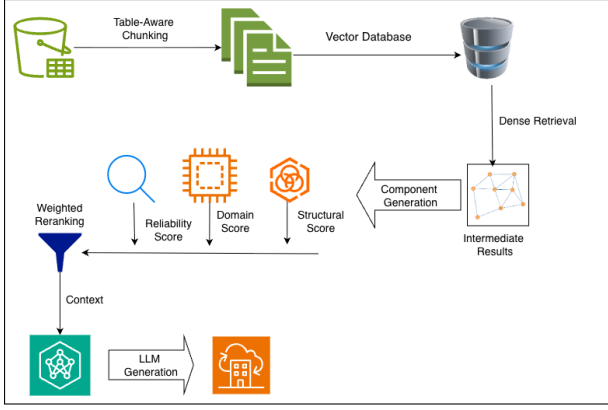


Figure 2: **End-to-end system.** STaB-RAG pipeline: tables are chunked and stored in a vector database, dense retrieval produces intermediate results, components are scored (reliability, domain, structural), and weighted reranking selects context for LLM generation.

tation fields. Each table is reconstructed as a structured object, preserving headers, rows, and schema-level metadata following table-parsing methods such as TAPAS (Herzig et al., 2020) and TaBERT (Yin et al., 2020).

To enable standardized retrieval evaluation, we convert this curated corpus into the BEIR format (Thakur et al., 2021), generating: (i) `corpus.jsonl` with chunked table representations; (ii) `queries.jsonl` with all table-referenced questions; and (iii) `qrels.tsv` linking each query to its relevant table chunks. This BEIR-style split serves as the foundation for the subsequent chunking and retrieval experiments.

2.2 Phase 2: Evaluation on Multi-Granular Chunks

Chunking divides semi-structured tables into retrievable units. Motivated by hierarchical and semantic chunking strategies in recent RAG systems (Cheerla, 2025), we evaluate several granularities:

- **table_row:** Each row becomes an independent chunk, supporting record-level questions but losing cross-row structure.
- **table_column:** Columns are chunked individually, useful for attribute-focused queries.
- **pure_table:** A full-table serialization that preserves headers, row order, and global structure (Herzig et al., 2020; Yin et al., 2020).
- **sliding_window:** Token-based windows over the linearized table (Beltagy et al., 2020).

```
{
  "id": "{table_id}_pure_table",
  "table": {
    "columns": ["Column1", "Column2", "Column3"],
    "rows": [
      ["value1", "value2", "value3"],
      ["value4", "value5", "value6"],
      ["value7", "value8", "value9"]
    ]
  },
  "chunk_type": "pure_table",
  "table_id": "unique_table_identifier"
}
```

Figure 3: **pure_table chunk format.** Each table is stored as a single chunk. The table field preserves the full structure via columns and rows. The `chunk_type` field is set to "pure_table" to distinguish this representation from other chunking strategies, and `table_id` links the chunk back to the underlying source table.

While effective for long-document retrieval, this underperformed compared to **pure_table** for structured table QA.

Across all models and rerankers tested, the **pure_table** representation yielded the most stable and robust retrieval performance.

2.3 Phase 3: Embedding and Reranking Exploration

To understand which retrieval signals and encoders work best for table QA, we built an experimental multi-stage retrieval stack and plugged in different models and scoring functions.

Retrieval stack. Given a query, the pipeline proceeds as follows:

- **Dense retrieval.** A dense encoder retrieves a top- k set of semantically related table chunks, following standard practices in dense passage retrieval (Karpukhin et al., 2020).
- **BM25 lexical scoring.** For these candidates, we compute a BM25 score (Robertson et al., 2009) to capture exact token overlap between the query and the (chunked) table text.
- **Structural similarity scoring.** We add a structural score that measures how well the query aligns with table layout cues (e.g., headers, column groupings), motivated by schema-aware table modeling (Yin et al., 2020; Deng et al., 2022).

- **Weighted re-ranking.** The dense, lexical, and structural scores are combined into a single weighted score used to re-rank candidate table chunks.
- **Cross-encoder re-ranking (optional).** In some configurations, a cross-encoder (Reimers and Gurevych, 2019) is applied on top of the weighted score to further refine the ordering.

Embedding model comparison- We instantiated this stack with several embedding models:

- Sentence-Transformers variants,
- RoBERTa-based encoders,
- Qwen Embedding models.

Across these experiments, Qwen3-Embedding-4B (Zhang et al., 2025) produced the most stable and informative representations for table-structured text, consistently outperforming the alternatives. We therefore adopt it as the primary encoder in our final pipeline described in the next phase.

2.4 Phase 4: STaB Retriever Implementation

Phase 3 gave us the building blocks (dense, lexical, and structural signals) and identified Qwen3-Embedding-4B as the most reliable encoder. In Phase 4, we fix these design choices and instantiate our final Structure-Text-and-BM25 (STaB) retriever over the pure_table corpus. The final weighted reranking score is defined in Eq. 1.

$$\text{score}_{\text{rerank}} = \alpha \cdot \text{Domain} + \beta \cdot \text{Reliability} + \gamma \cdot \text{Structure}. \quad (1)$$

Concretely, for each query STaB proceeds as follows:

1. **Candidate generation (text).** Use Qwen3-Embedding-4B to retrieve a top- k set of semantically relevant table chunks.
2. **Signal attachment.** For each candidate, attach:
 - a dense similarity score (from Qwen embeddings),
 - a BM25 score (lexical overlap),
 - a structural similarity score (alignment with the table schema/layout).
3. **Weighted aggregation.** Combine the three signals into a single weighted score and rank candidates accordingly.

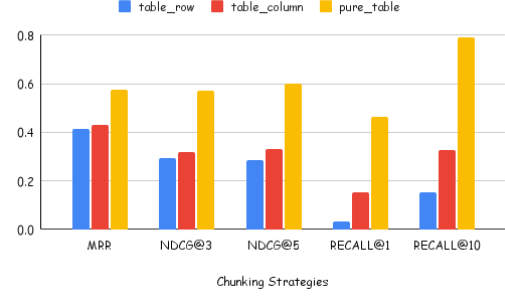


Figure 4: Model performance on different chunking strategies

3 Results Obtained from the Chosen Approach

As shown in Table 1, STaB-RAG outperforms all baseline models across the majority of metrics, achieving the highest NDCG@3 and Recall@1 scores. While models like SSDR_{im} and BIBERT perform competitively on NDCG@5 and Recall@10, STaB-RAG delivers the strongest early-precision performance, particularly in Recall@1, a key indicator of dense retrieval quality. This improvement highlights the effectiveness of combining lexical, structural, and dense semantic signals, allowing STaB-RAG to retrieve top-relevant table segments more reliably than any individual model.

4 Analysis of Results and Findings

4.1 Chunking Analysis:

We evaluated three chunking strategies, table_row, table_column, and pure_table, using the Qwen3-Embedding-4B model. Figure 4 illustrates the model’s performance across five metrics: MRR, NDCG@3, NDCG@5, Recall@1, and Recall@10 (Li et al., 2025). Across all evaluated metrics, the pure_table strategy consistently outperforms both table_row and table_column. The difference is especially pronounced for Recall@1 and Recall@10, where pure_table achieves substantially higher scores. These results indicate that Qwen3-Embedding-4B benefits from preserving full-table structure during chunking, leading to more effective retrieval performance compared to row-based or column-based segmentation.

4.2 Study of Components in the Weighted Reranking:

To understand the contribution of each component within our weighted reranking mechanism, We conducted a step-wise evaluation across five metrics:

Table 1: Results on NQ-Tables dataset.

Model	NDCG@3	NDCG@5	Recall@1	Recall@10
BM25	23.59	25.52	17.95	37.65
DTR	44.70	49.13	31.72	70.03
SPLADE	54.59	58.46	38.73	83.13
TAPAS	56.93	61.08	43.91	<u>85.95</u>
BIBERT	57.53	<u>61.73</u>	44.69	84.87
SSDR _{im}	<u>58.50</u>	61.89	<u>45.14</u>	86.71
STaB-RAG	58.66	61.57	48.68	80.35

Bold and underline indicate the best and sub-optimal performance respectively.

MRR, NDCG@3, NDCG@5, Recall@1, and Recall@10 (Li et al., 2025). As shown in Figure 5, the baseline Dense Retriever provides the initial performance reference across all metrics.

In the second stage, we augment the dense retrieval results with a structural embedding score, capturing table-level structural similarity between queries and candidate chunks. This combined Dense Search + Structural Score approach yields consistent improvements across all metrics, particularly benefiting Recall@1 and NDCG scores. The improvement arises because structural signals help prioritize candidate chunks that are organizationally aligned with the query intent, something dense embeddings alone may overlook.

Finally, we evaluate our proposed STaB-RAG framework, which integrates BM25 lexical matching, structural similarity, and dense retrieval scores within a unified weighted reranking scheme. STaB-RAG achieves the highest overall performance, reinforcing the complementary nature of lexical, structural, and dense semantic signals. The most notable gains are observed in Recall@1 and Recall@10, demonstrating enhanced robustness in identifying the most relevant table segments.

4.3 Performance comparison of different models with STaB-RAG:

Table 1 summarizes the performance of several retrieval models on the NQ-Tables dataset across four metrics: NDCG@3, NDCG@5, Recall@1, and Recall@10 (Li et al., 2025). Among all compared systems, STaB-RAG achieves the best overall performance, with the highest scores highlighted in bold and sub-optimal scores are underlined.

A key observation is that Recall@1, the most critical metric for dense retrieval, is substantially im-

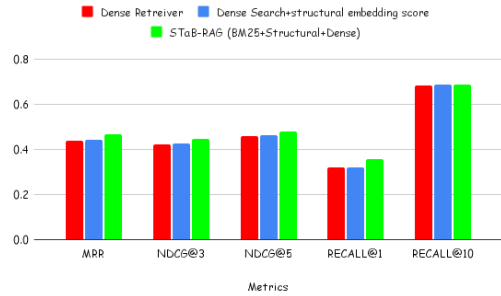


Figure 5: Ablation study of the components in the Weighted Reranking

proved by STaB-RAG. While traditional dense retrievers such as BIBERT and SPLADE rely exclusively on semantic similarity, their Recall@1 values remain limited. In contrast, STaB-RAG achieves a Recall@1 of 48.68, outperforming all baselines. This improvement stems from the integration of lexical evidence (BM25) and structural similarity signals, which help identify table regions that are highly relevant but may not be captured by dense embeddings alone.

STaB-RAG also delivers competitive performance on the ranking-based metrics. Its NDCG@3 (58.66) is the highest among all models, indicating stronger prioritization of relevant chunks at the top of the ranking. For NDCG@5, STaB-RAG remains near-optimal, closely matching the best-performing baseline. Similarly, its Recall@10 (80.35) remains strong, demonstrating robust retrieval coverage even beyond the top result.

Overall, these results show that STaB-RAG’s combination of lexical, structural, and dense semantic signals produces a more reliable and context-aware retrieval ranking, especially in scenarios where dense embeddings alone are insufficient. The sub-

stantial gains in Recall@1, supported by competitive performance across other metrics, validate the effectiveness of our weighted reranking design for table-based retrieval tasks.

5 Conclusions

Overall, the results demonstrate that STaB-RAG consistently outperforms all baseline dense, lexical, and structure-aware retrievers, particularly on Recall@1, the most critical metric for dense retrieval. By integrating complementary lexical, structural, and semantic signals within a unified weighted reranking framework, STaB-RAG is able to more reliably surface the most relevant table segments that dense embeddings alone often miss. This synergy leads to stronger early-precision performance while maintaining competitive results across all ranking metrics, highlighting the effectiveness of our proposed approach for table-based retrieval tasks.

6 Contributions

Amit Ashok Manjarly – Conducted an extensive review of state-of-the-art Retrieval-Augmented Generation (RAG) methods, with a particular emphasis on table-aware and hierarchical chunking strategies. Led the effort in identifying appropriate evaluation metrics for retrieval and QA performance, and aligned them with BEIR and NQ-Table benchmarks.

Atharva Gupta – Analyzed existing open-source RAG implementations and underlying architectures through GitHub repositories. Adapted and extended these pipelines to support the proposed WeightedRAG framework, including configuring multi-stage retrieval and embedding backends for table-specific use cases.

Animesh Kumar – Performed literature research focused on Graph-RAG, table-structured information retrieval, and graph-based reasoning. Investigated approaches to preserve and exploit tabular relationships using graph neural networks and structured representations to inform the WeightedRAG graph-reranking module.

Sankarshana W. Nanjangud – Designed and formalized evaluation benchmarks for the Natural Questions (NQ-Table) dataset. Implemented the conversion of NQ tables into BEIR-style format, implemented cache logic to store the vector

embeddings and validated retrieval metrics across multiple chunking and embedding configurations.

Shreyas Ramani – Established evaluation protocols for BEIR-style table datasets, with emphasis on consistent preprocessing, chunk filtering, and qrels generation. Assisted in running large-scale retrieval experiments and comparing dense, sparse, and hybrid strategies for benchmarking WeightedRAG.

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